

HLD & System Design Practice Roadmap

Phases, question list, topics covered, and 6-week practice plan

Use this roadmap to move from HLD foundations to product-scale and distributed-system questions. Complete each phase in order, and for every question follow the same answer structure given near the end of this PDF.

Phase 1: Foundation Design Questions

No.	Question	Main Topics Covered
1	Design URL Shortener like Bitly	API design, DB schema, cache, scaling, redirects
2	Design Rate Limiter	Token bucket, Redis, API Gateway, distributed system
3	Design Pastebin	Storage, expiry, DB, cache, public/private access
4	Design File Upload Service	Object storage, metadata DB, CDN, upload flow
5	Design Image Upload Service	S3/object storage, CDN, compression, async processing
6	Design Notification System	Kafka, email/SMS/push, retry, DLQ
7	Design Logging System	Kafka, Elasticsearch/OpenSearch, storage, search
8	Design API Gateway	Routing, auth, rate limiting, load balancing
9	Design Authentication System	Login, JWT, sessions, OAuth, Redis
10	Design Distributed Cache	Consistent hashing, replication, eviction

Topics covered: HTTP, API design, Load balancing, Caching, Rate limiting, Object storage, Kafka, Redis, DB schema, Basic scaling

Phase 2: Intermediate Product Systems

No.	Question	Main Topics Covered
11	Design WhatsApp 1-to-1 Chat	WebSocket, message delivery, online status
12	Design Group Chat	Fanout, message ordering, Kafka, storage
13	Design Instagram Feed	Feed generation, caching, ranking, fanout
14	Design Twitter/X Timeline	Read-heavy system, fanout on write/read
15	Design YouTube Video Upload	Video processing, transcoding, CDN
16	Design Netflix	Video streaming, CDN, recommendations basics
17	Design Google Drive	File storage, sharing, permissions, versioning
18	Design Dropbox Sync	File sync, conflict resolution, metadata
19	Design Search Autocomplete	Trie, Redis, Elasticsearch/OpenSearch, ranking
20	Design E-commerce Product Search	Elasticsearch/OpenSearch, filters, indexing pipeline

Topics covered: WebSocket, CDN, Search, Ranking, Async processing, Large file storage, Feed design, Eventual consistency, Fanout, Permissions

Phase 3: Transaction and Consistency Heavy Systems

No.	Question	Main Topics Covered
21	Design BookMyShow	Seat locking, concurrency, DB

		transactions
22	Design Payment System	Idempotency, consistency, retries, audit logs
23	Design Wallet System	Double-entry ledger, strong consistency
24	Design UPI-like Payment Flow	Distributed transaction, timeout, reconciliation
25	Design Food Delivery App	Order flow, location, payment, notification
26	Design Uber/Ola	Location tracking, matching, pricing
27	Design Inventory Management	Stock deduction, race condition, consistency
28	Design Coupon System	Abuse prevention, locking, rate limiting
29	Design Ticket Booking System	Seat/slot booking, locking, payment flow
30	Design Banking Transaction System	ACID, ledger, audit, failure handling

Topics covered: Transactions, Concurrency, Distributed locks, Idempotency, Saga pattern, Outbox pattern, Race conditions, Strong consistency, Reconciliation, Audit logs

Phase 4: Scale Heavy Systems

No.	Question	Main Topics Covered
31	Design YouTube	Upload, transcoding, recommendation, streaming
32	Design Instagram	Feed, media upload, stories, likes
33	Design Twitter/X	Timeline, tweet, follow, trending
34	Design LinkedIn Feed	Feed, connections, ranking
35	Design Facebook News Feed	Fanout, ranking, privacy
36	Design Google Search	Crawling, indexing, ranking
37	Design Gmail	Email receive/send, search, storage
38	Design WhatsApp	Chat, media, groups, delivery status
39	Design Slack/Discord	Channels, real-time chat, notification
40	Design Zoom	Video call, signaling, media server basics

Topics covered: Massive scale, Partitioning, Sharding, CDN, Search indexing, Real-time systems, Low latency, High availability, Feed ranking, Storage optimization

Phase 5: Infrastructure/System Component Designs

No.	Question	Main Topics Covered
41	Design Load Balancer	L4/L7, health check, algorithms
42	Design Consistent Hashing System	Ring, virtual nodes, scaling
43	Design Distributed Queue	Kafka-like basics, partitions, consumer groups
44	Design Job Scheduler	Cron, retries, distributed locking
45	Design Web Crawler	Queue, dedupe, rate limit, indexing
46	Design Metrics Monitoring System	Logs, metrics, alerts, p95/p99
47	Design Feature Flag System	Config, rollout, targeting
48	Design Config Management System	Dynamic config, versioning, rollback
49	Design Distributed Lock	Redis/Zookeeper, expiry, fencing token
50	Design Leaderboard System	Redis sorted set, ranking, scale

Topics covered: Internal infra, Distributed systems, Leader election, Scheduling, Monitoring, Queues, Locks, Consistency

Best 30 Questions for Java Backend Interviews

No.	Question	No.	Question
1	URL Shortener	16	Google Drive
2	Rate Limiter	17	E-commerce Product Search

3	Notification System	18	BookMyShow
4	API Gateway	19	Payment System
5	Authentication System	20	Wallet System
6	File Upload Service	21	Food Delivery App
7	Image Upload Service	22	Uber/Ola
8	Distributed Cache	23	Inventory Management
9	Logging System	24	Coupon System
10	Search Autocomplete	25	Distributed Queue
11	WhatsApp 1-to-1 Chat	26	Job Scheduler
12	Group Chat	27	Web Crawler
13	Instagram Feed	28	Metrics Monitoring System
14	Twitter Timeline	29	Distributed Lock
15	YouTube Video Upload	30	Leaderboard System

How to Practice Each Question

1. Functional requirements
2. Non-functional requirements
3. Capacity estimation
4. API design
5. Database schema
6. High-level architecture
7. Component deep dive
8. Scaling strategy
9. Caching strategy
10. Database partitioning/sharding
11. Async processing/Kafka
12. Failure handling
13. Security
14. Monitoring
15. Final trade-offs

6-Week Practice Plan

Week 1: Basics

- Day 1: URL Shortener
- Day 2: Rate Limiter
- Day 3: Pastebin
- Day 4: File Upload Service
- Day 5: Notification System
- Day 6: API Gateway
- Day 7: Revision + diagrams

Week 2: Backend Core

- Day 8: Authentication System
- Day 9: Distributed Cache
- Day 10: Logging System

- Day 11: Search Autocomplete
- Day 12: E-commerce Product Search
- Day 13: Job Scheduler
- Day 14: Revision

Week 3: Real-time Systems

- Day 15: WhatsApp 1-to-1 Chat
- Day 16: Group Chat
- Day 17: Slack/Discord
- Day 18: Notification deep dive
- Day 19: WebSocket scaling
- Day 20: Message ordering
- Day 21: Revision

Week 4: Feed and Media Systems

- Day 22: Instagram Feed
- Day 23: Twitter Timeline
- Day 24: YouTube Video Upload
- Day 25: Netflix Streaming
- Day 26: Google Drive
- Day 27: Dropbox Sync
- Day 28: Revision

Week 5: Transactions

- Day 29: BookMyShow
- Day 30: Payment System
- Day 31: Wallet System
- Day 32: Inventory Management
- Day 33: Coupon System
- Day 34: Food Delivery App
- Day 35: Revision

Week 6: Advanced Scale

- Day 36: Uber/Ola
- Day 37: Web Crawler
- Day 38: Metrics Monitoring System

- Day 39: Distributed Queue
- Day 40: Distributed Lock
- Day 41: Leaderboard System
- Day 42: Final revision

Recommended Starting Order

1. URL Shortener
2. Rate Limiter
3. Notification System
4. File Upload Service
5. Authentication System
6. Distributed Cache
7. Search Autocomplete
8. WhatsApp Chat
9. Instagram Feed
10. BookMyShow
11. Payment System
12. Uber

Java / Spring Boot Angle for Every Design

For each system design, connect HLD to implementation by answering these questions:

Which service?
Which controller?
Which DB table?
Which cache key?
Which Kafka topic?
Which failure case?
Which monitoring metric?

Tip: After completing one design, revise it once with numbers: QPS, storage, cache hit ratio, DB partitions, Kafka partitions, p95/p99 latency, failure handling, and monitoring metrics.